Web Data Integration Project Proposal (Group 6) – Revised

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Abstract. We integrate book data from Goodreads (52,478 records), DBpedia (59,270) and Amazon (40,000) into a common schema. Following the feedback, the entity is a book *edition* (e.g. the 2018 Putnam hardcover of *Where the Crawdads Sing*), not a work. Using the ISBN only as an evaluation key, 6,087 editions occur in at least two datasets. Many works appear in several editions with the same title and author (6,588 works with 33,540 editions in the Amazon subset), which makes telling editions apart the central challenge for identity resolution. The integrated schema has 14 attributes, 12 of which occur in at least two datasets; `authors` and `genres` are list attributes.

1 Use Case and Entity Definition

The goal is to integrate book information from three independently maintained sources that describe overlapping books with different attribute names, formats and levels of detail. One record of the integrated dataset describes one **edition**: a specific publication of a work by a publisher, in a format (hardcover, paperback, ...), at a date, with its own page count and ISBN. Goodreads and Amazon list editions directly. A DBpedia resource describes a work, but its infobox values (ISBN, publisher, release date, pages) are those of the first edition, so each DBpedia record is treated as the first edition of its work. Matching therefore has to distinguish, e.g., the hardcover, paperback, large-print and movie tie-in editions of the same novel, which share title and author but differ in publisher, date, format and page count.

2 Datasets

Goodreads. The "Best Books Ever" dataset (scraped from the Goodreads list of the same name) lists one edition per book. We use the original CSV; an earlier copy had been re-saved with a spreadsheet program, which had turned 91% of the ISBNs into scientific notation and altered dates.

DBpedia. All 59,270 resources of type `dbo:Book` were downloaded from the public SPARQL endpoint. Because the endpoint silently cuts off long-running queries, the download (1) lists the book URIs in partitions by

first character, paged with `LIMIT 10000 OFFSET` (below the 40,000 offset limit), and (2) fetches the attributes for 150 known URIs per query (`VALUES`); a block that returns a partial result (HTTP 206) is split and fetched again. The final count equals `COUNT(dbo:Book)`. Year and language combine `dbo:releaseDate`/`dbo:language` with the raw infobox properties `dbp:releaseDate`, `dbp:pubDate` and `dbp:language`, which lowers missing values from 88.5% to 11.4% (year) and from 87.6% to 20.4% (language). Unlike the first version, books without an ISBN are no longer excluded, so 39.0% of the records have no ISBN.

Amazon. The metadata file of Amazon Reviews 2023 (McAuley Lab) contains 4,448,181 book products. 4,336,308 have a title and an author, and 3,526,297 of these also have an ISBN; this is the population we sample from. Products without ISBN (mostly Kindle and Audible) are excluded because the ISBN is our gold-standard key. From this population we take the **40,000 editions with the most ratings** (at least 6,506 ratings each). The selection uses only Amazon's own rating counts, no information from the other datasets. We chose popularity because Goodreads "Best Books Ever" is itself a popularity-based list: a uniform random sample of 40,000 editions (fixed seed) shares only 331 editions with Goodreads, compared with 2,169 for the popularity-based subset (Table 3). Amazon reports one rating count for all editions of a work, so the selection contains popular works together with many of their editions – exactly the hard cases for edition-level matching. This choice has a cost. The overlap figures in Table 3 are higher than they would be for a random Amazon sample. They describe this subset and should not be generalized to all of Amazon. Our aim is a benchmark with enough shared editions, not a representative sample.

Attributes with more than 30% missing values are marked (MV). In Goodreads these are `series` (55.3%), `characters` (73.8%), `edition` (90.6%), `firstPublishDate` (40.6%), `awards` (79.8%) and `setting` (77.9%); in DBpedia `pages` (38.4%), `genres` (36.4%) and `isbns` (39.0%); in Amazon only `edition` (40.1%).

3 Integrated Schema

Table 2 maps the source attributes to the integrated schema. Compared with the first proposal, `format` and `edition` were added because they distinguish editions of the same work, and the publication date is kept per edition instead of being reduced to a year.

Twelve attributes occur in at least two datasets and seven in all three (`title`, `authors`, `genres`, `publication_date`, `publisher`, `page_count`, `language`), which allows fusion by voting. Representations differ: Amazon titles carry series or edition details in brackets ("Where the Crawdads Sing (Movie Tie-in)"), Goodreads stores contributors and roles in one string ("J.K. Rowling, Mary GrandPré (Illustrator)"), DBpedia dates range from years to full dates, and the sources use different genre vocabularies.

Table 1. Overview of the datasets.

Dataset	Source	Format	# Ent.	# Attr.	List of attributes
Goodreads Best Books Ever	Kaggle / Zenodo[1]	CSV	52,478	25	bookId, title, series (MV), author, rating, description, language, isbn, genres, characters (MV), bookFormat, edition (MV), pages, publisher, publishDate, firstPublishDate (MV), awards (MV), numRatings, ratingsByStars, likedPercent, setting (MV), coverImg, bbeScore, bbeVotes, price
DBpedia	SPARQL endpoint[2]	XML	59,270	9[4]	title, authors, publisher, release_date, year, pages (MV), language, genres (MV), isbns (MV)
Amazon Reviews 2023	Hugging Face[3]	CSV	40,000	14[4]	title, authors, format, publisher, edition (MV), publication_date, language, pages, genres, average_rating, ratings_count, price, isbn13, isbn10

[1] `https://www.kaggle.com/datasets/arnabchaki/goodreads-best-books-ever`, original: `https://doi.org/10.5281/zenodo.4265096`
[2] `https://dbpedia.org/sparql`, script `dbpedia_books.py`
[3] `https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023`, script `amazon_subset.py`
[4] Without the identifiers `uri` (DBpedia) and `amazon_id`, `asin` (Amazon).

Table 2. Attribute intersection with the integrated schema (one record per edition).

Attribute name	Type	Datasets in which the attribute is found
title	string	GR (title), DB (title), AM (title)
authors	list<string>	GR (author), DB (authors), AM (authors)
genres	list<string>	GR (genres), DB (genres), AM (genres)
publication_date	date	GR (publishDate), DB (release_date), AM (publication_date)
publisher	string	GR (publisher), DB (publisher), AM (publisher)
page_count	integer	GR (pages), DB (pages), AM (pages)
language	string	GR (language), DB (language), AM (language)
format	string	GR (bookFormat), AM (format)
edition	string	GR (edition), AM (edition)
rating	float	GR (rating), AM (average_rating)
ratings_count	integer	GR (numRatings), AM (ratings_count)
price	float	GR (price), AM (price)
series	string	GR (series)
awards	list<string>	GR (awards)
Identifier for overlap estimation and the gold standard only (not used for matching):		
isbn	string	GR (isbn), DB (isbns), AM (isbn13, isbn10)

GR = Goodreads, DB = DBpedia, AM = Amazon; source column names in brackets.

4 Overlap Estimation

Two records describe the same edition if they share an ISBN (ISBN-10 and ISBN-13 normalised to ISBN-13). The ISBN is used only for this estimate and as the gold standard; the matchers in Phase II will not use it. Since records without an ISBN cannot be counted, the numbers in Table 3 are lower bounds.

Table 3. Edition overlap (shared ISBN-13) and comparison with a random Amazon sample.

Measure	**Editions**
Amazon ∩ Goodreads	2,169
Amazon ∩ DBpedia	1,725
Goodreads ∩ DBpedia	3,271
In all three datasets	539
In at least two datasets	**6,087**
Random Amazon sample (40,000, seed 42) ∩ Goodreads	331

With 6,087 editions in at least two datasets, the requirement of 1,000 is met even under this strict criterion. The total number of distinct editions is well above 2,500, since the three sources contain 151,748 records with at most 6,087 shared editions.

To show the difficulty of edition-level matching, we also compared works (normalised title and first-author surname). Goodreads and DBpedia share 10,619 works; for 4,759 of the 7,561 shared works with ISBNs on both sides (63%), the two sources list *different* editions. Amazon and Goodreads share 3,146 works (22% in different editions), and within the Amazon subset 6,588 works appear in more than one edition (33,540 records). A matcher based on title and author alone would therefore produce many false matches; date, publisher, format and page count are needed to tell editions apart. In Phase II, the matchers will therefore compare title, authors, publisher, publication date, format and page count. Title and authors first generate candidate pairs. The other attributes then decide whether two candidates are the same edition. We evaluate against the ISBN gold standard.